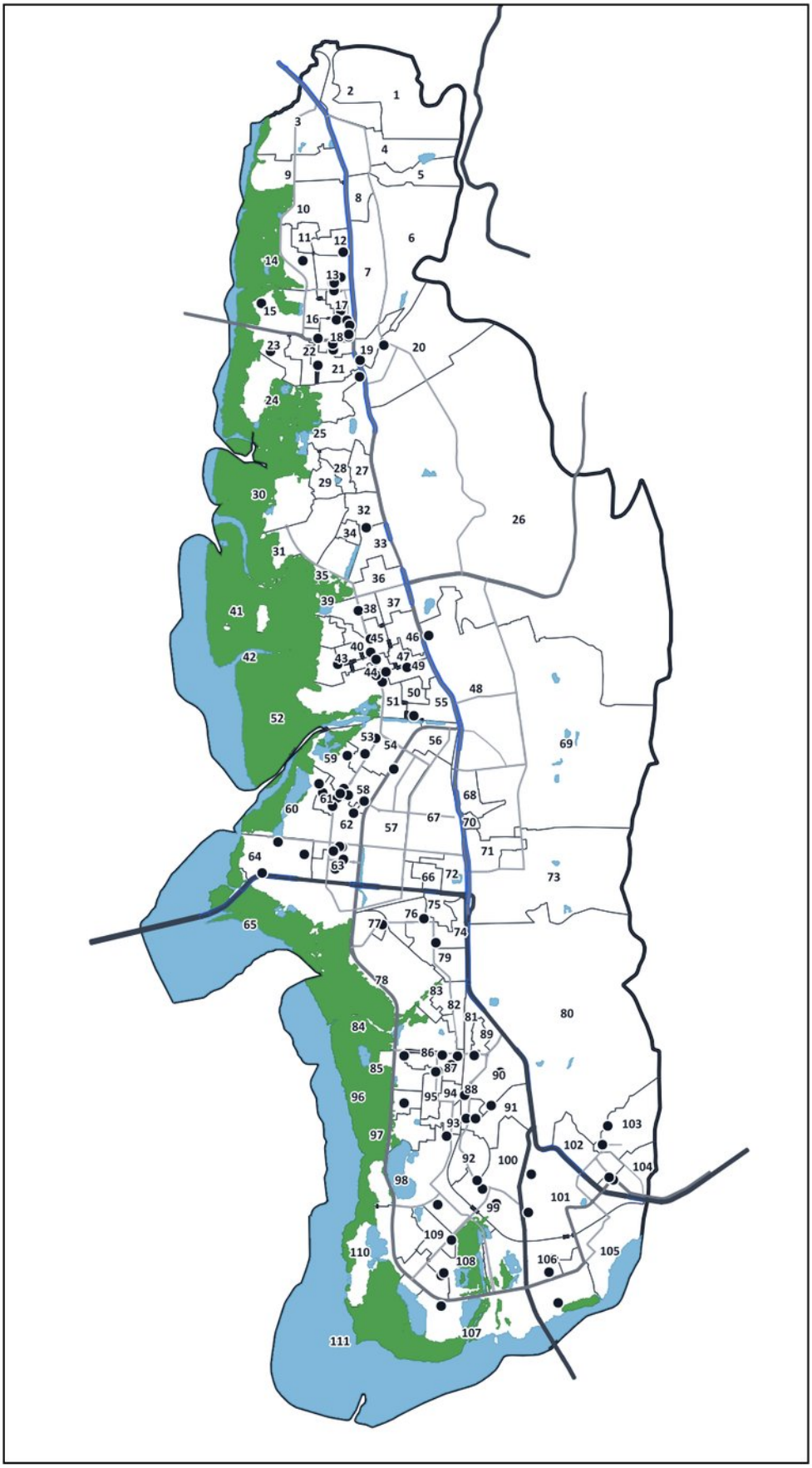
Study area and ward boundaries

Navi Mumbai: 111 wards, major roads, water and mangrove



Legend

- Wards (111) with ward numbers

Major roads (OSM)

- Waterlogged 26 May 2025
- Motorway / trunk
- Primary
- Secondary

Water / mangrove

- Water
- Mangrove
- Wetland
- Hospitals and clinics (OSM)

KEY NUMBERS

- 111 wards, 122.9 km² in total
- Ward numbers run from north (1) to south (111)
- Roads: OpenStreetMap
- Water: WorldCover 2021

HOW TO READ THIS MAP

Each shape is one ward and the number is its Ward_ID. Blue road lines were reported waterlogged on 26 May 2025. Green and blue areas on the west side are mangrove and creek water.

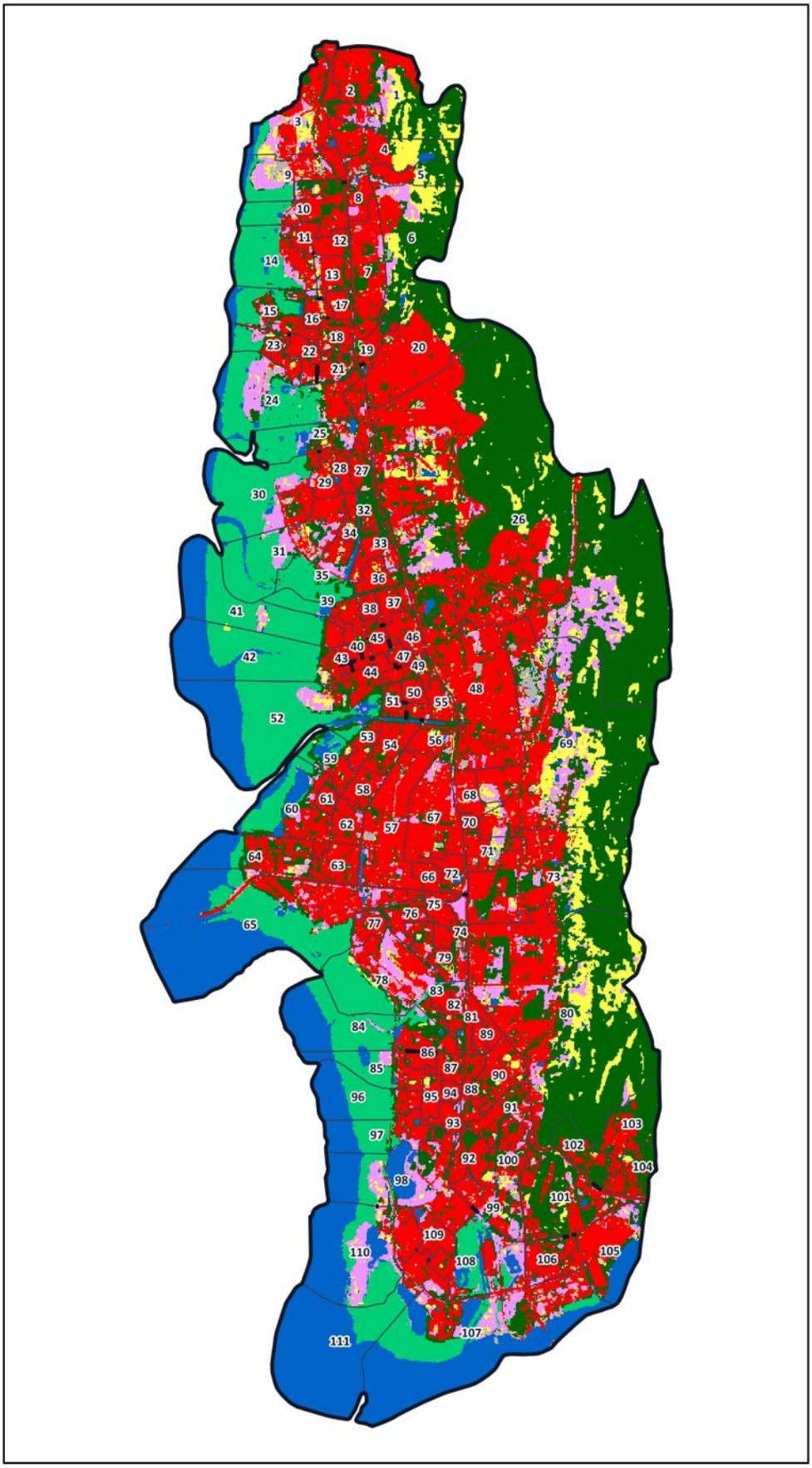
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Land cover and built-up land

ESA WorldCover 2021 (10 m)



Legend

Land cover (WorldCover)

- Tree cover
- Shrubland
- Grassland
- Cropland
- Built-up
- Bare / sparse
- Water
- Wetland
- Mangrove

KEY NUMBERS

- Built-up: 33.1%
- Tree cover: 26.4%
- Mangrove: 13.8%
- Water: 13.3%
- Cropland: 7.0%
- Grass: 5.2%

HOW TO READ THIS MAP

Red is built-up land. Green is trees and mangrove. Blue is water. Most built-up land sits in the central strip of the city.

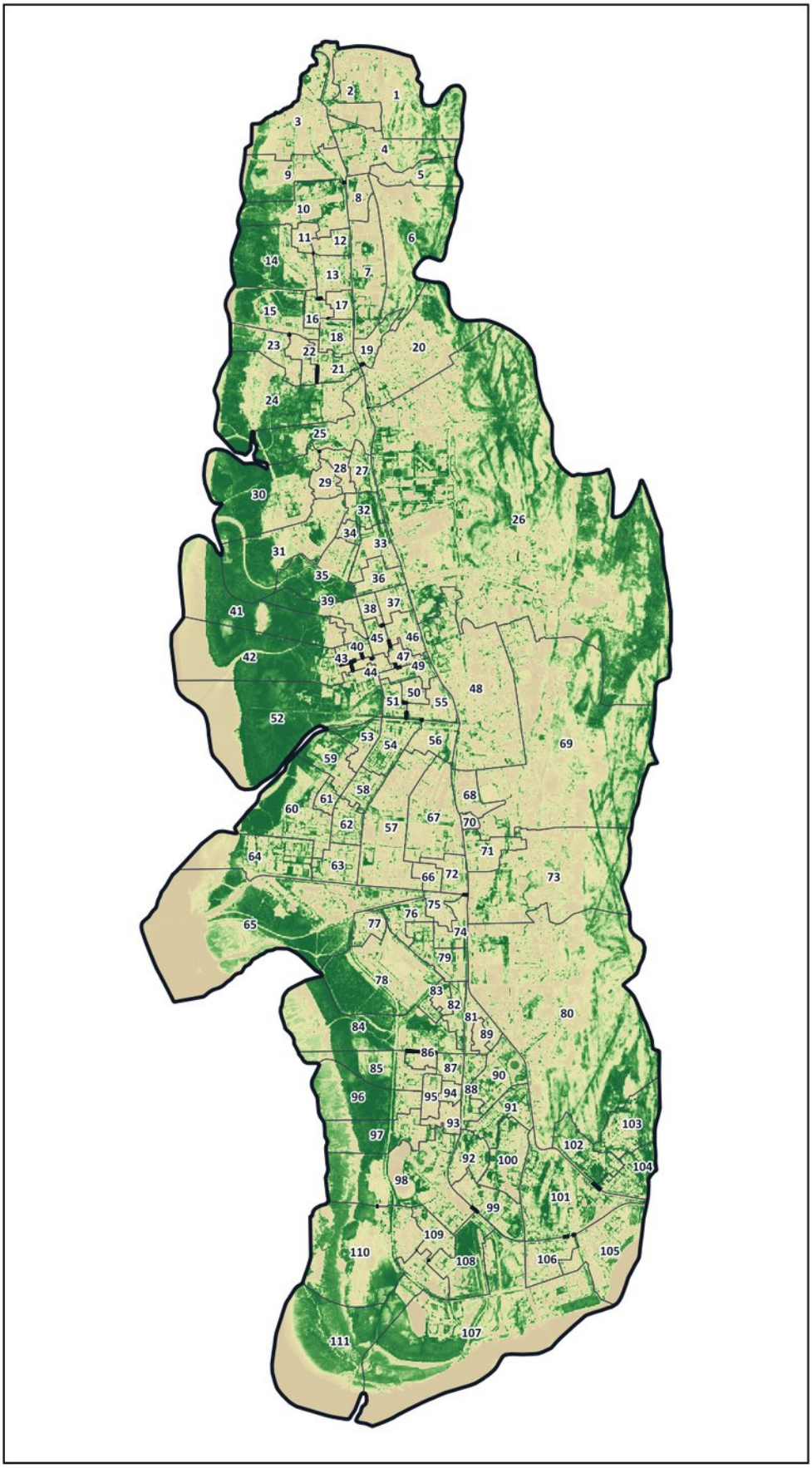
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Vegetation (NDVI)

Sentinel-2, median of 8 and 11 April 2026



Legend

NDVI (greenness)

0.7 or more (dense plants)
0 or less (bare / built-up)

KEY NUMBERS

- City mean NDVI: 0.30
- Ward means: 0.12 to 0.49
- 40% of the area is below 0.2
- Mangroves stay green in the west

HOW TO READ THIS MAP

NDVI measures how green the ground is. Dark green means dense plants. Pale means bare or built-up ground. Low-NDVI wards get a high green-cover deficit score.

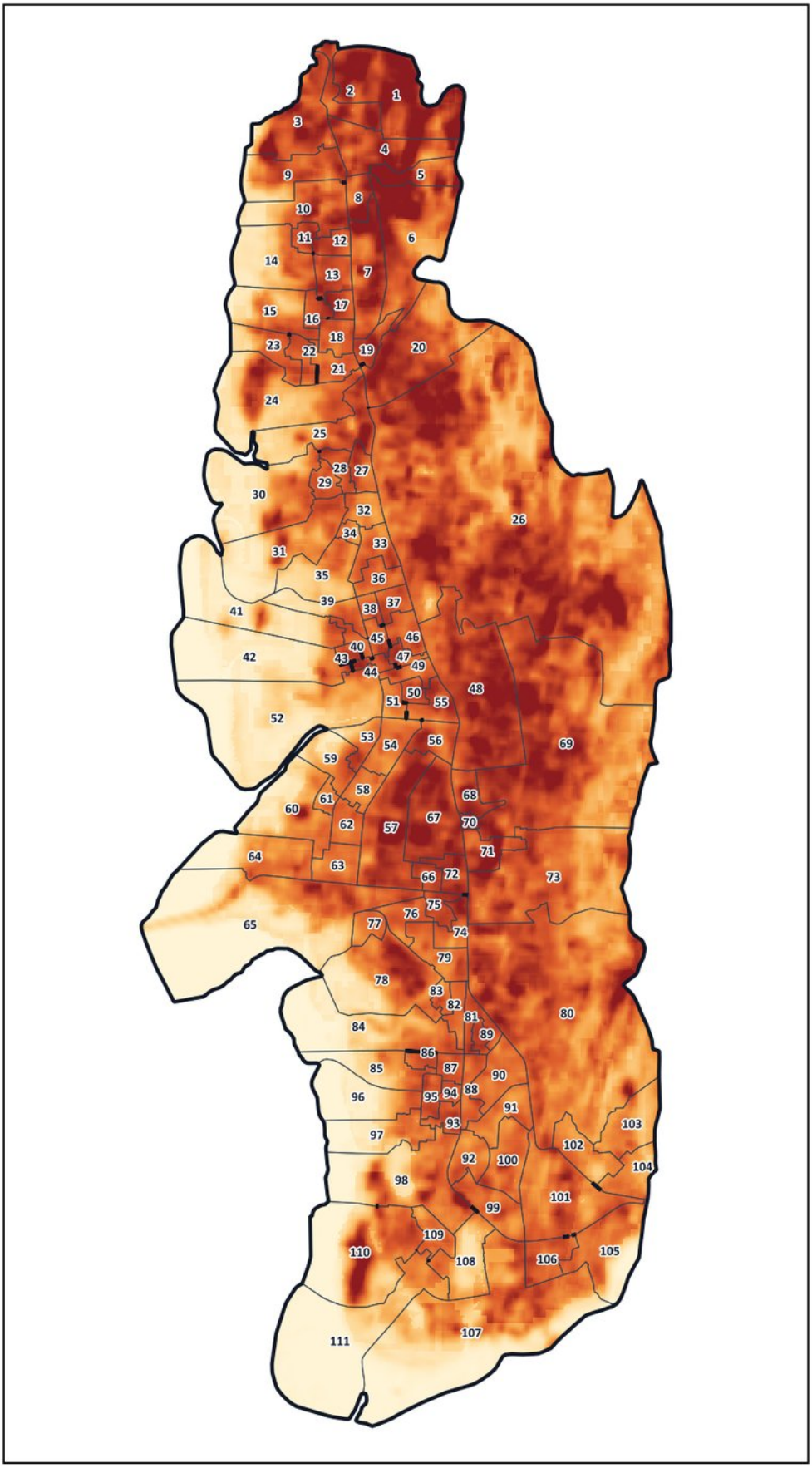
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Land surface temperature (heat)

Landsat 8/9, median of 13, 22 and 30 April 2026



Legend

Surface temperature

48 °C (hot)
32 °C or less (cool)

KEY NUMBERS

- City mean: 41.0 °C
- Ward means: 33.7 to 47.6 °C
- 23.0% of the area is 45 °C or hotter
- Hottest ward: ward 70 (47.6 °C)

HOW TO READ THIS MAP

Darker means a hotter ground surface. This is the temperature of roofs, roads and soil seen by the satellite. It is higher than the air temperature people feel.

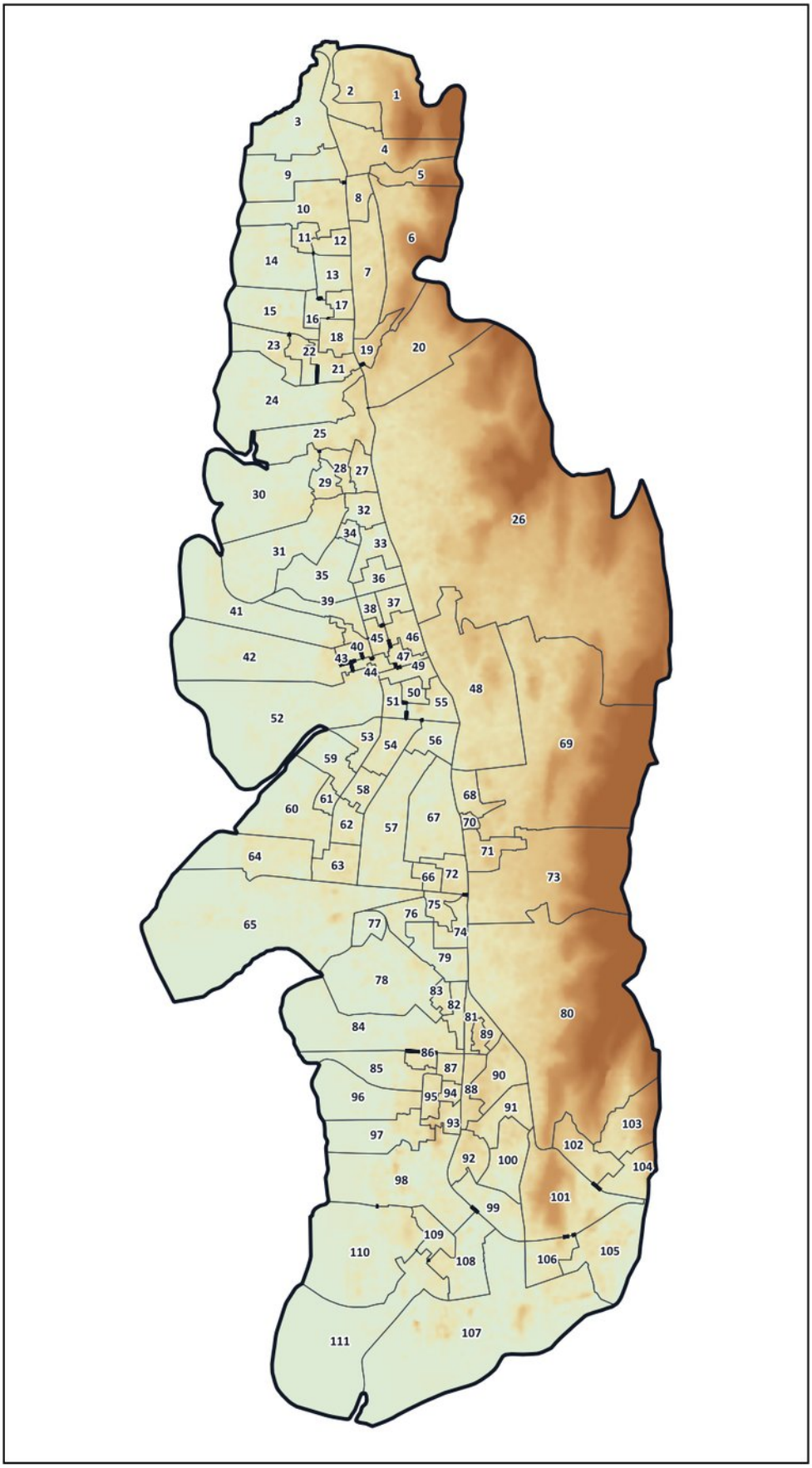
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Elevation

SRTM, 30 m



Legend

Elevation (m)

200 m or more
0 m or less

KEY NUMBERS

- Low ground along the creek in the west
- Higher ground in the east (Parsik hills)
- Elevation is the top flood factor (43% weight)

HOW TO READ THIS MAP

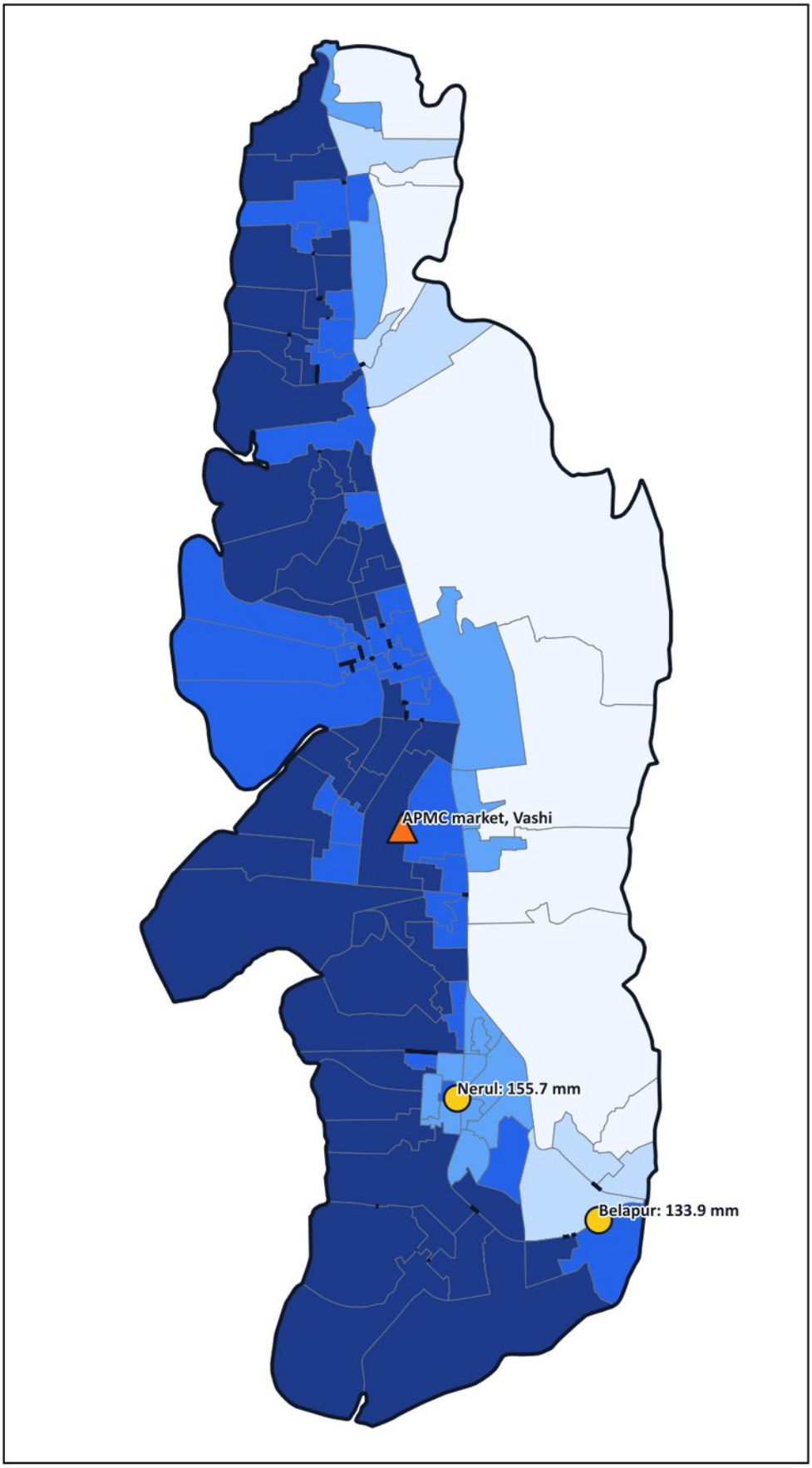
Pale colours are low ground. Brown is higher ground. Low ground collects water more easily. SRTM includes roofs and trees, so it is a surface height.

0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Flood susceptibility

Ward score from elevation, water distance, slope and built-up land (AHP weights 43 / 27 / 16 / 14)



Legend

Flood susceptibility (1-5)

- 1.95 - 2.26
- 2.26 - 3.1
- 3.1 - 3.72
- 3.72 - 4.3
- 4.3 - 4.84

Flood reports, 26 May 2025

- Rain record
- Waterlogged site

KEY NUMBERS

- A susceptibility index, not a forecast
- AHP consistency ratio: 0.017 (below 0.10 is acceptable)
- 26 May 2025: Nerul 155.7 mm, Belapur 133.9 mm (news report)

HOW TO READ THIS MAP

Darker blue means water can collect more easily. Points show the 26 May 2025 news reports (reported rainfall for Nerul and Belapur, and the waterlogged APMC market). Point positions are approximate.

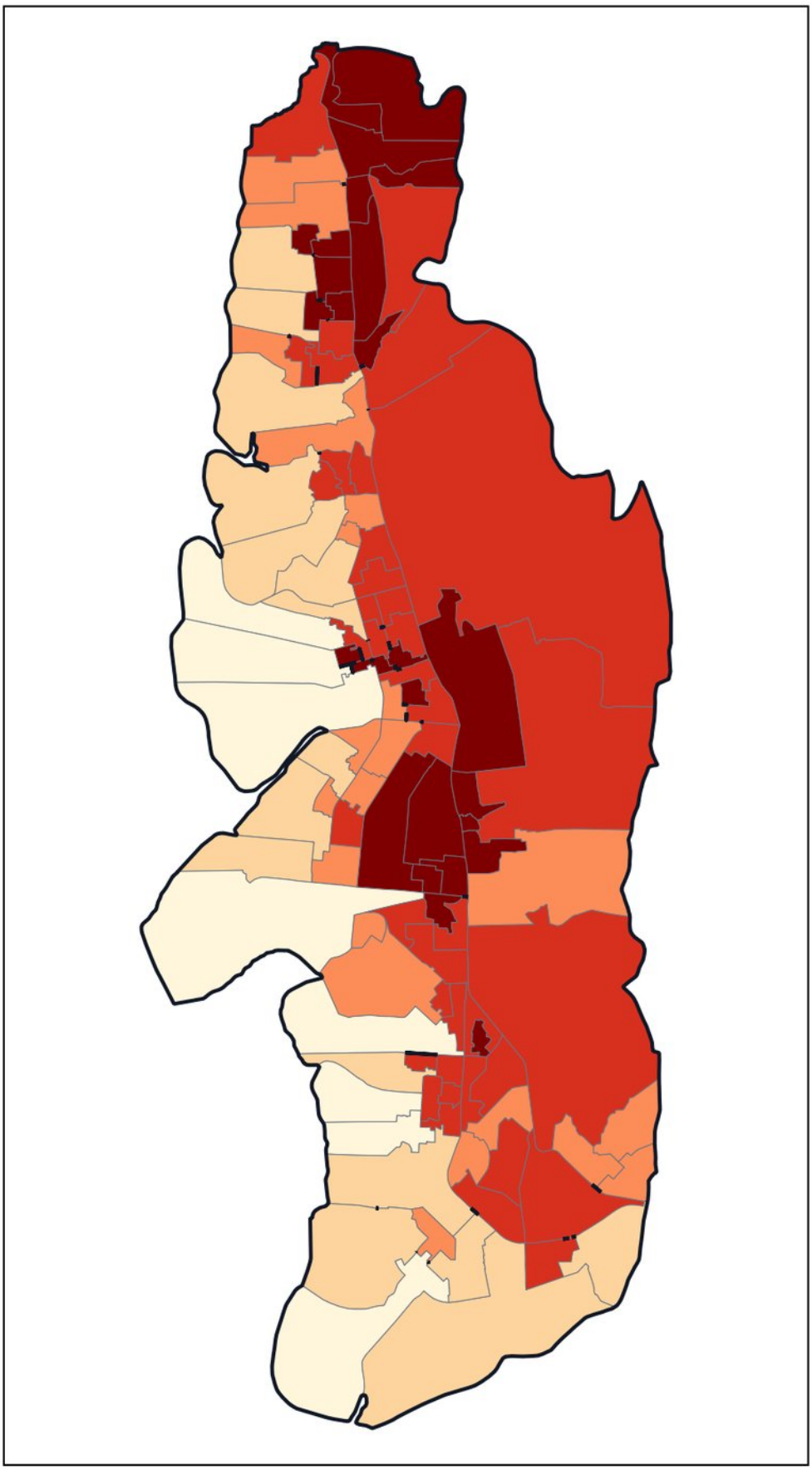
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery. Flood reports: Free Press Journal, 26 May 2025.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Heat stress score

Ward temperature stretched to a 1 (coolest) to 5 (hottest) scale



Legend

Heat stress score (1-5)

- 1 - 1.76
- 1.76 - 2.7
- 2.7 - 3.58
- 3.58 - 4.17
- 4.17 - 5

KEY NUMBERS

- Score 5 = hottest ward
- Score 1 = coolest ward
- Median of 3 April 2026 scenes

HOW TO READ THIS MAP

Each ward gets one score from its average surface temperature. Dark red wards are the hottest. Classes use Natural Breaks (Jenks).

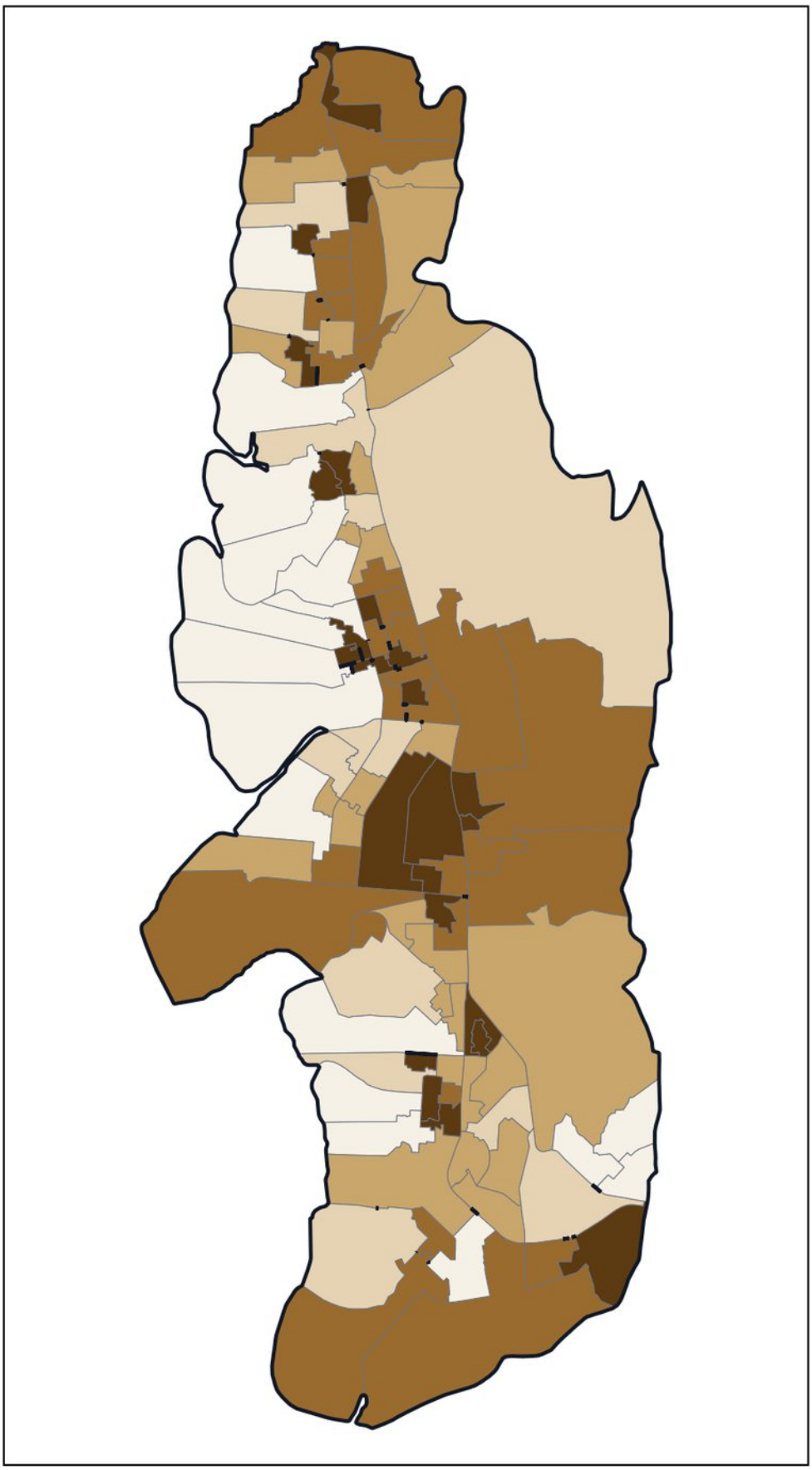
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Green-cover deficit score

Ward NDVI turned upside down: fewest plants = 5



Legend

Green deficit score (1-5)

- 1 - 1.88
- 1.88 - 2.76
- 2.76 - 3.48
- 3.48 - 4.09
- 4.09 - 5

KEY NUMBERS

- Score 5 = least vegetation
- Score 1 = most vegetation
- One season only (April 2026)

HOW TO READ THIS MAP

Dark brown wards have the fewest plants. NDVI is reversed so that a higher score always means more stress. Classes use Natural Breaks (Jenks).

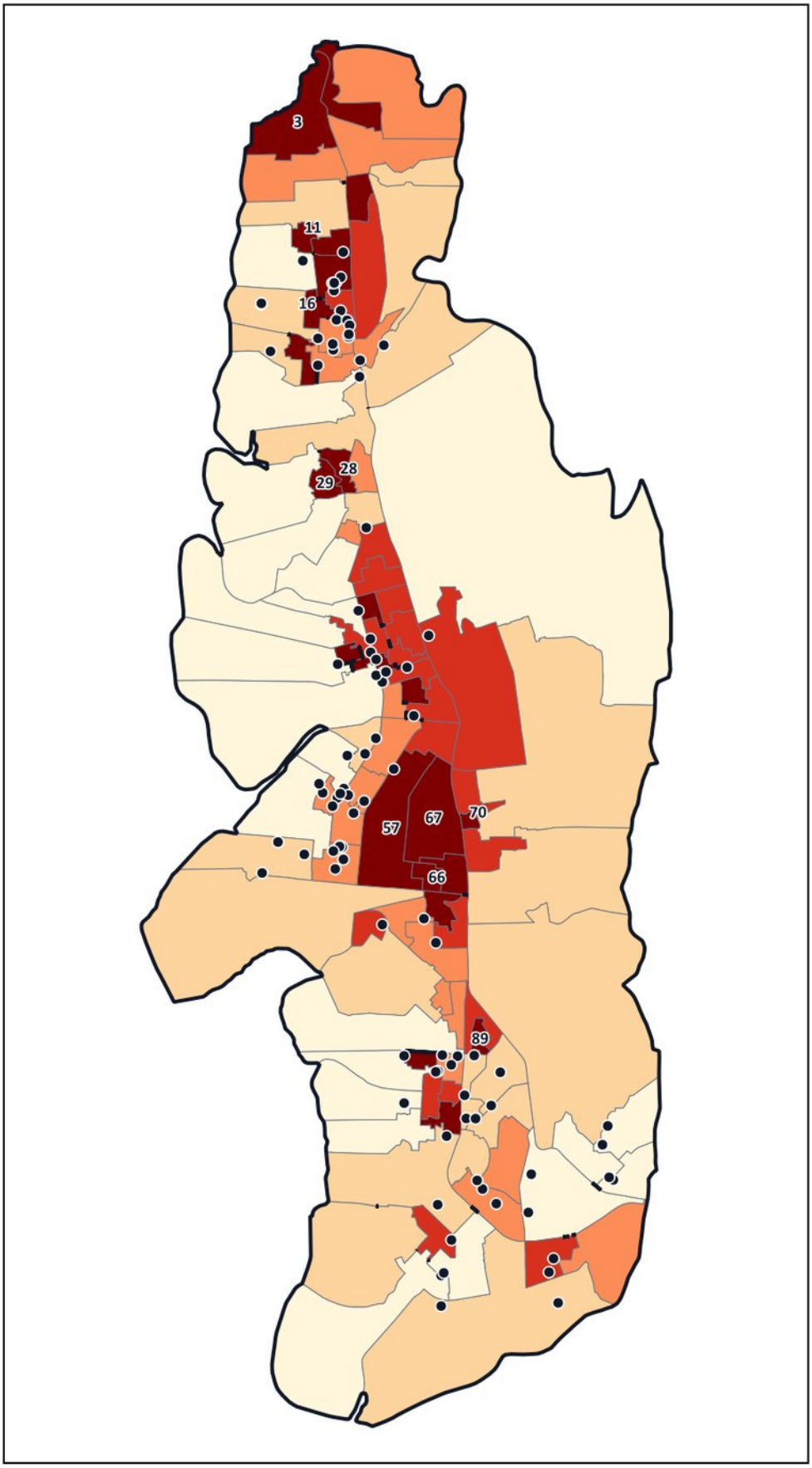
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Urban Climate Stress Index (UCSI)

UCSI = (Heat + Green-cover deficit + Flood) ÷ 3. Five classes of about 22 wards (quantiles)



Legend

UCSI class

- Highest
- High
- Medium
- Low
- Lowest
- Hospitals and clinics (OSM)

KEY NUMBERS

- Highest 5 wards: 70, 57, 67, 11, 29
- Highest class has 23 wards (two wards tie)
- Wards 57 and 66 are in the top quarter for all three stresses
- Rankings hold when weights change (rank correlation 0.98 and 0.95)

HOW TO READ THIS MAP

Dark red wards have the highest combined stress. Numbers mark the 10 highest-ranked wards. Black dots are hospitals and clinics. UCSI compares wards. It does not predict a future event.

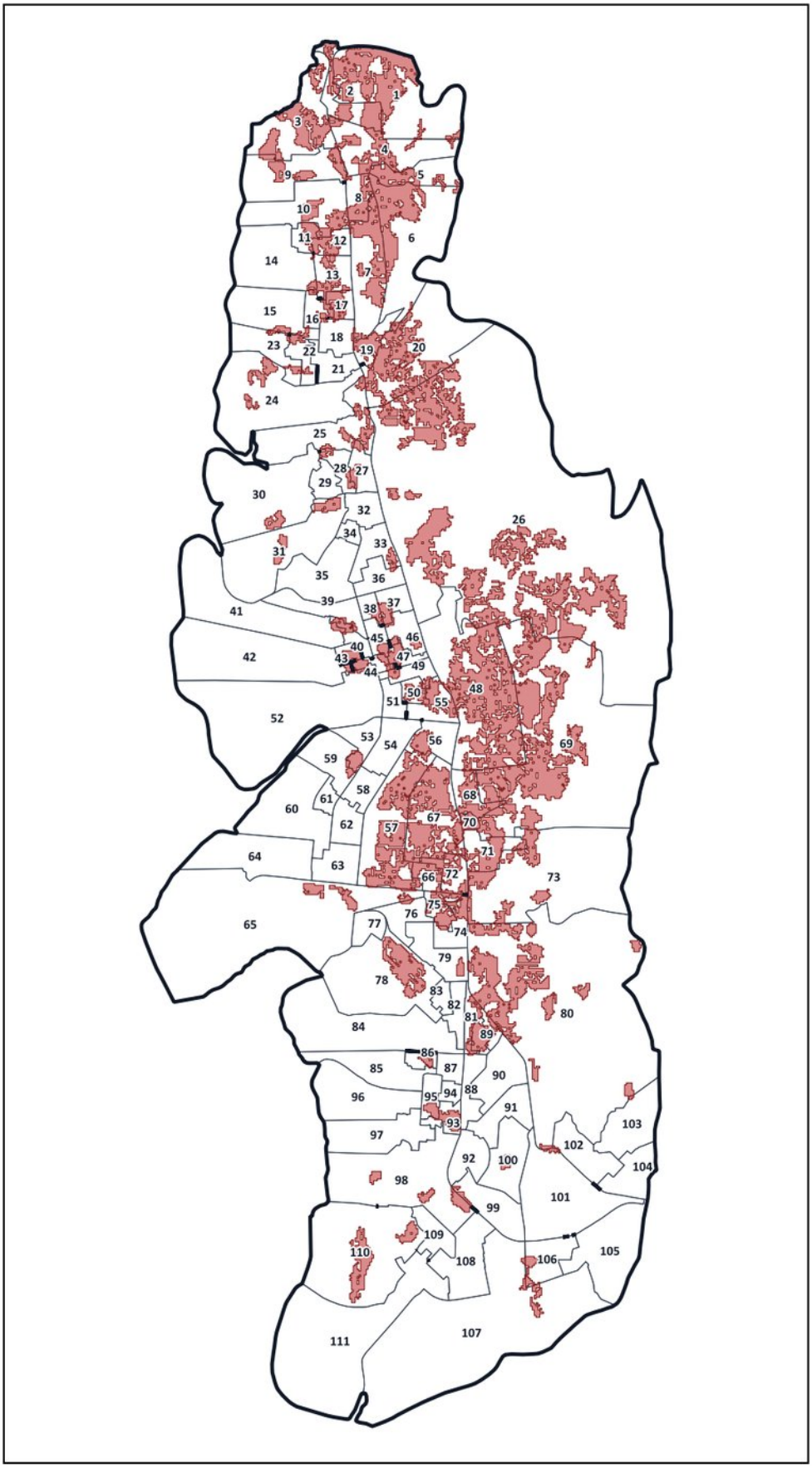
0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | MGM University | CRS: WGS 84 / UTM zone 43N (EPSG:32643)

Hot and low-green patches

Areas at 45 °C or more AND NDVI below 0.2 (spatial overlay)



Legend

Hot and low-green patches

KEY NUMBERS

- All hot and low-green pixels: 18.9 km² (15.4% of the area)
- Patches of 2 ha or more (mapped): 71 patches, 17.6 km²

HOW TO READ THIS MAP

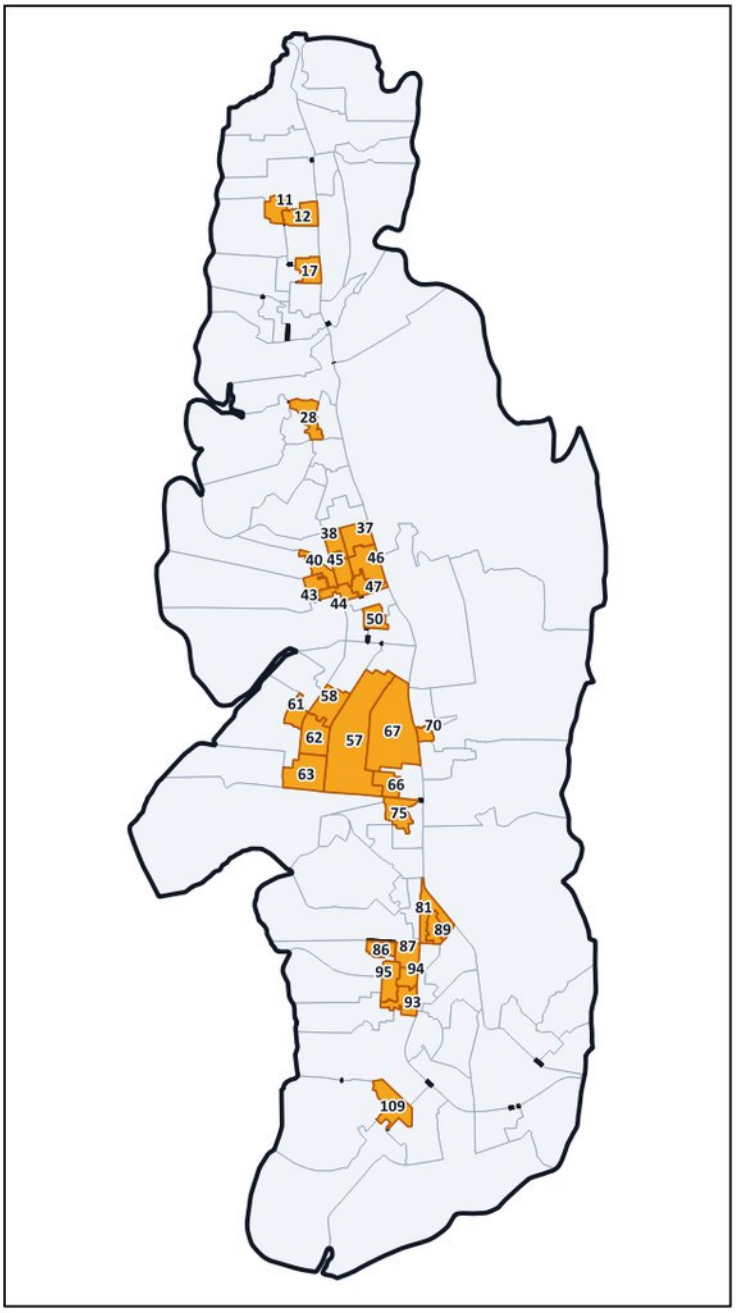
Each red patch is hot ground with very few plants. These are the first places to check for tree planting or cool roofs.

0 2 4 km

Data: Landsat 8/9 Collection 2 Level 2 (USGS/NASA), Sentinel-2 L2A (ESA), ESA WorldCover 2021, SRTM (NASA/USGS), OpenStreetMap contributors. Ward boundaries: OpenCity / BharatLAS (not an official NMMC file). April 2026 imagery.

Query A1: Built-up land 80% or more

30 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"BuiltUp_Pct" >= 80

QGIS tool: Processing > Extract by expression
```

RESULT

30 of 111 wards match

WHAT IT MEANS

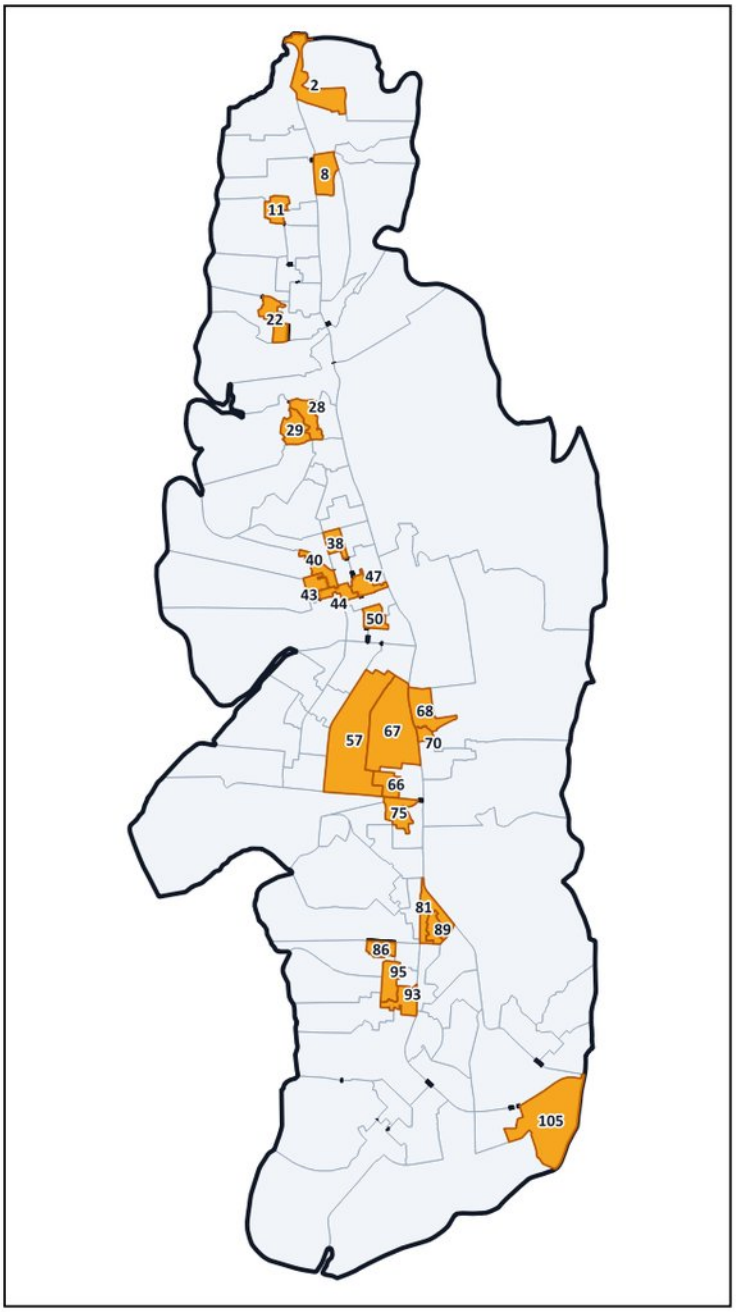
30 wards are 80% or more built-up. Ward 70 is the most built-up (98.25%).

Table: first 12 of 30 rows, sorted by Built-up %, highest first.

Ward_ID	BuiltUp_Pct	LST_C	UCSI
70	98.25	47.597	4.48
44	97.42	45.351	4.21
94	96.66	44.526	4.06
43	95.84	44.847	4.19
38	95.4	43.937	4.16
89	95.2	45.716	4.31
86	91.57	44.235	4.13
81	90.98	44.709	4.03
50	90.53	45.005	4.18
45	89.64	43.914	3.99
11	87.68	45.305	4.41
12	86.91	45.074	4.19

Query A2: Vegetation index (NDVI) below 0.2

24 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"NDVI" < 0.2

QGIS tool: Processing > Extract by expression
```

RESULT

24 of 111 wards match

WHAT IT MEANS

24 wards have NDVI below 0.2, which means very few plants. Ward 70 has the lowest value (0.1164).

Result of the query

Other wards

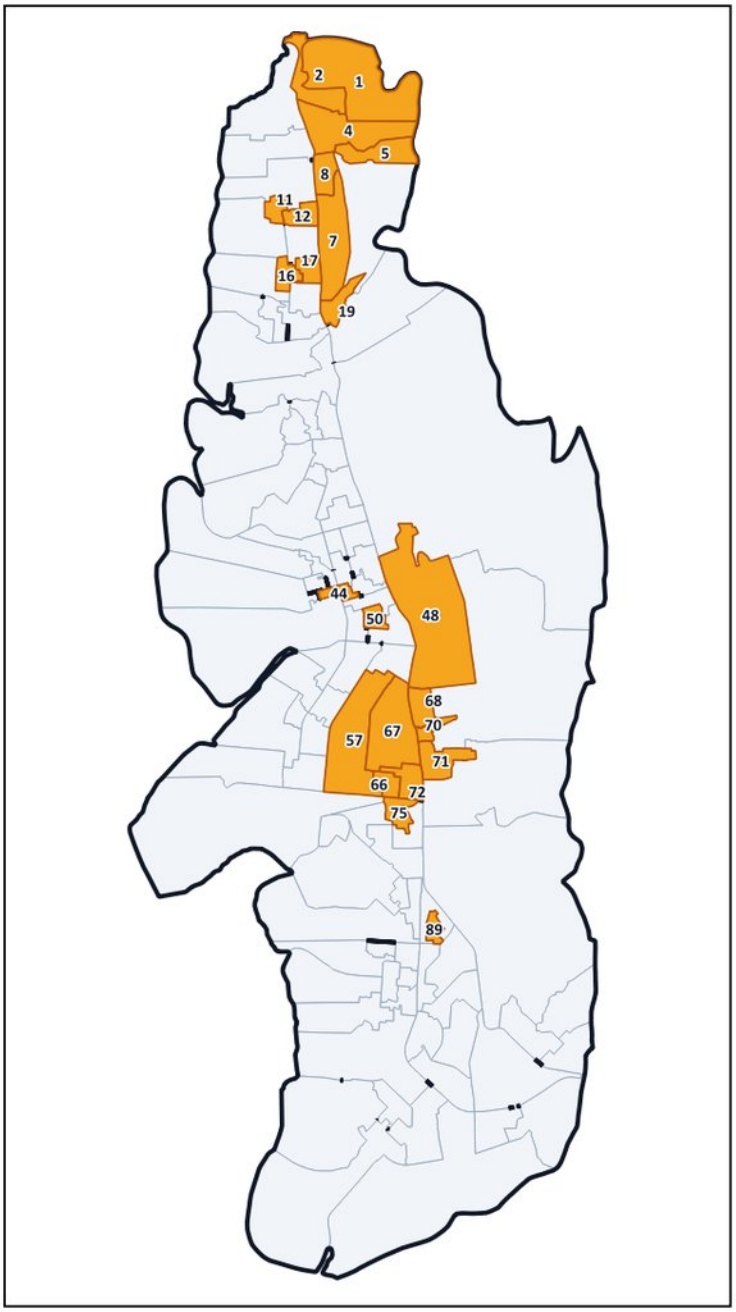
0 2 4 km

Table: first 12 of 24 rows, sorted by NDVI, lowest first.

Ward_ID	NDVI	GreenCover_Pct	UCSI
70	0.1164	1.6	4.48
89	0.1252	3.4	4.31
29	0.1369	11.9	4.4
11	0.1399	9.5	4.41
93	0.1429	6.2	4.14
68	0.1575	11.8	4.12
2	0.1664	23.7	4.15
66	0.17	6.9	4.39
44	0.1729	2.2	4.21
57	0.1761	8.4	4.48
105	0.1763	15.8	3.72
28	0.1769	11.5	4.3

Query A3: Surface temperature 45 °C or more

23 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"LST_C" >= 45

QGIS tool: Processing > Extract by expression
```

RESULT

23 of 111 wards match

WHAT IT MEANS

23 wards have an average ground surface temperature of 45 °C or more. Ward 70 is the hottest (47.6 °C). This is ground temperature, not air temperature.

Result of the query

Other wards

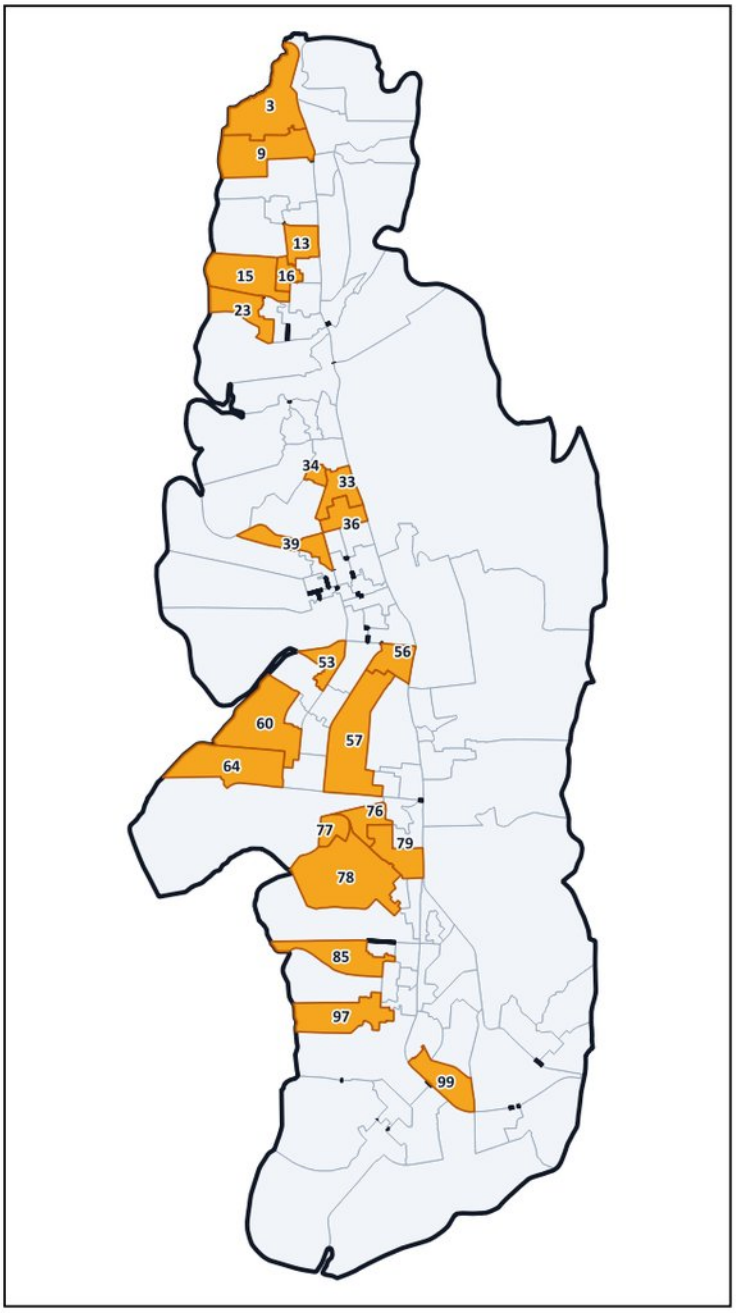
0 2 4 km

Table: first 12 of 23 rows, sorted by LST, hottest first.

Ward_ID	LST_C	NDVI	BuiltUp_Pct
70	47.597	0.1164	98.25
1	46.972	0.2418	31.73
67	46.442	0.1807	80.85
72	46.346	0.209	75.3
48	46.057	0.2018	78.83
5	45.983	0.261	30.45
71	45.959	0.2056	63.83
68	45.88	0.1575	74.6
57	45.847	0.1761	84.13
8	45.733	0.1838	70.96
89	45.716	0.1252	95.2
4	45.692	0.2424	46.3

Query A4: Flood susceptibility score 4.5 or more

22 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"Flood_Score" >= 4.5

QGIS tool: Processing > Extract by expression
```

RESULT

22 of 111 wards match

WHAT IT MEANS

22 wards score 4.5 or more out of 5. Susceptibility means water can collect there more easily. It is not a flood forecast.

Result of the query

Other wards

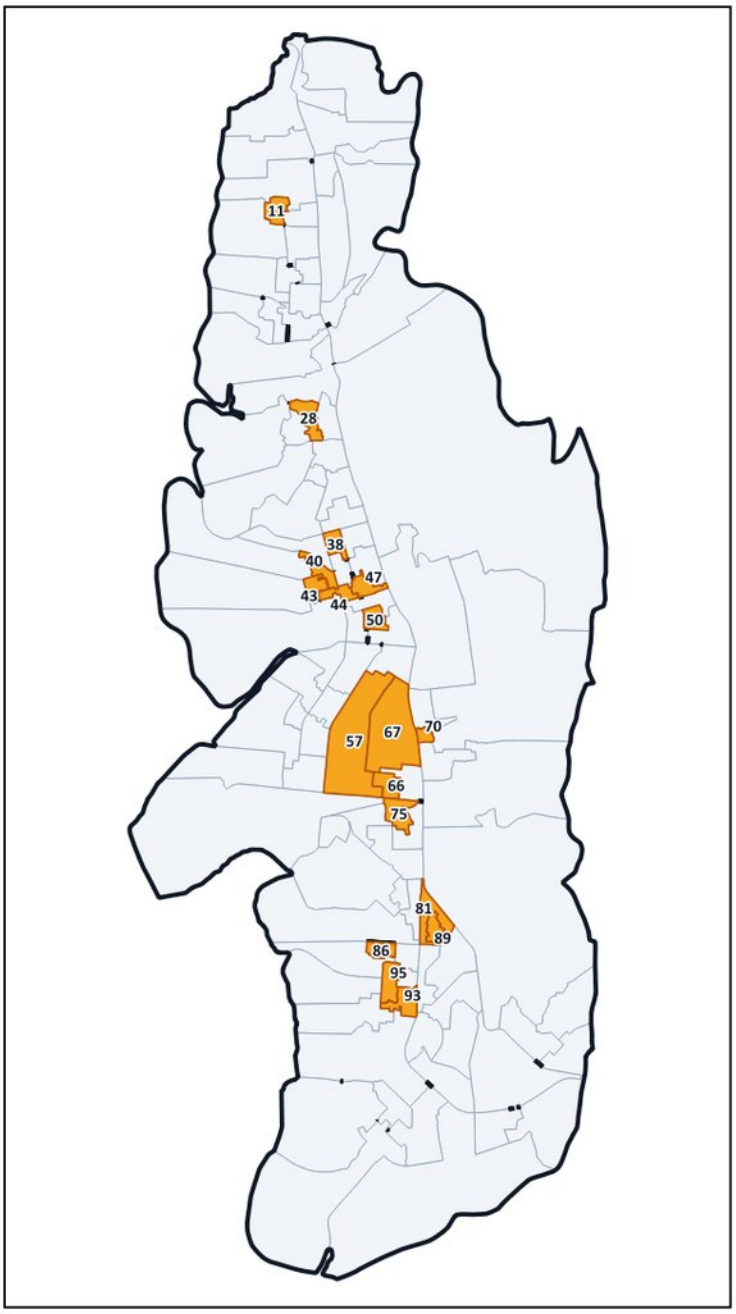
0 2 4 km

Table: first 12 of 22 rows, sorted by flood score, highest first.

Ward_ID	Flood_Score	Elev_m	DistWater_m
16	4.84	4.94	143
33	4.84	4.04	191.5
77	4.84	1.87	147.9
3	4.69	2.79	158.2
53	4.69	4.24	110.6
56	4.69	4.57	166
79	4.69	4.1	165.8
99	4.69	4.27	222.6
13	4.57	4.95	325.4
36	4.57	4.12	303.3
57	4.57	4.88	285.6
76	4.57	4.68	408

Query A5: Built-up 80% or more AND NDVI below 0.2

18 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"BuiltUp_Pct" >= 80 AND "NDVI" < 0.2

QGIS tool: Processing > Extract by expression
```

RESULT

18 of 111 wards match

WHAT IT MEANS

18 wards are both very built-up and very bare of plants. Ward 70 ranks highest in this group (UCSI rank 1 of 111).

Result of the query

Other wards

0 2 4 km

Table: first 12 of 18 rows, sorted by UCSI rank (1 = highest stress).

Ward_ID	BuiltUp_Pct	NDVI	LST_C	UCSI	UCSI_Rank
70	98.25	0.1164	47.597	4.48	1
57	84.13	0.1761	45.847	4.48	2
67	80.85	0.1807	46.442	4.43	3
11	87.68	0.1399	45.305	4.41	4
66	86.84	0.17	45.302	4.39	6
89	95.2	0.1252	45.716	4.31	7
28	84.01	0.1769	44.592	4.3	8
75	86.71	0.1864	45.565	4.27	11
44	97.42	0.1729	45.351	4.21	14
43	95.84	0.1888	44.847	4.19	16
50	90.53	0.1968	45.005	4.18	19
38	95.4	0.1982	43.937	4.16	20

Query S1: Wards intersecting water or mangrove (1 ha or more)

48 of 111 wards

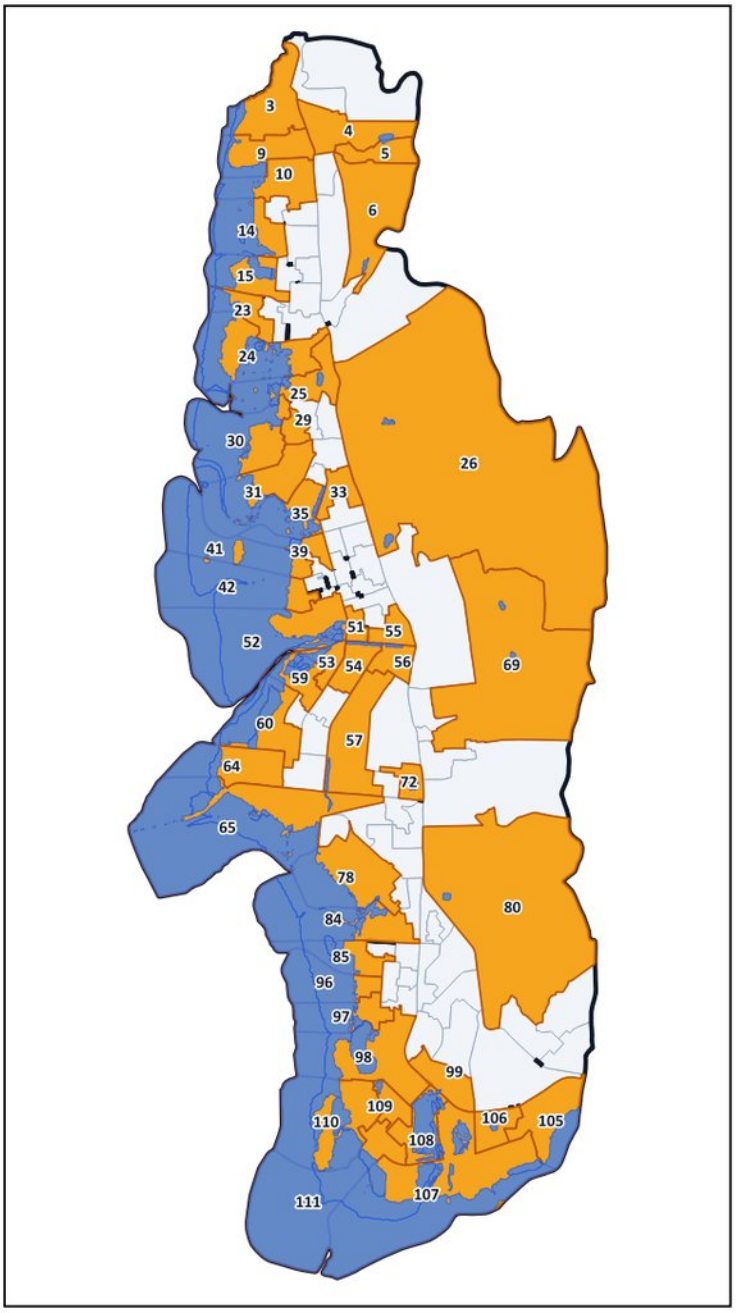


Table: first 12 of 48 rows, sorted by water share, highest first.

Ward_ID	Water_Pct	Flood_Score	UCSI
42	88.5	4.26	2.37
41	88.3	4.26	2.35
111	88.2	4.42	2.99
96	84.2	4.41	2.25
52	79.8	4.26	2.42
65	75.4	4.41	3.22
84	74.5	4.41	2.63
30	70.7	4.41	2.54
14	67.9	4.41	2.53
97	67.2	4.55	2.65
110	64.2	4.41	3.15
107	62.8	4.41	3.41

HOW IT WAS RUN IN QGIS

```
Vector > Research Tools > Select by Location
Input: wards
Predicate: intersect
Comparison: water_bodies where Area_ha >= 1
```

RESULT

48 of 111 wards

WHAT IT MEANS

48 wards touch a water or mangrove body of 1 ha or more (blue). Distance to water is one of the four flood factors, so these wards are the first to check.

Result of the query

Other wards

0 2 4 km

Query S2: Wards within 200 m of a major road

78 of 111 wards

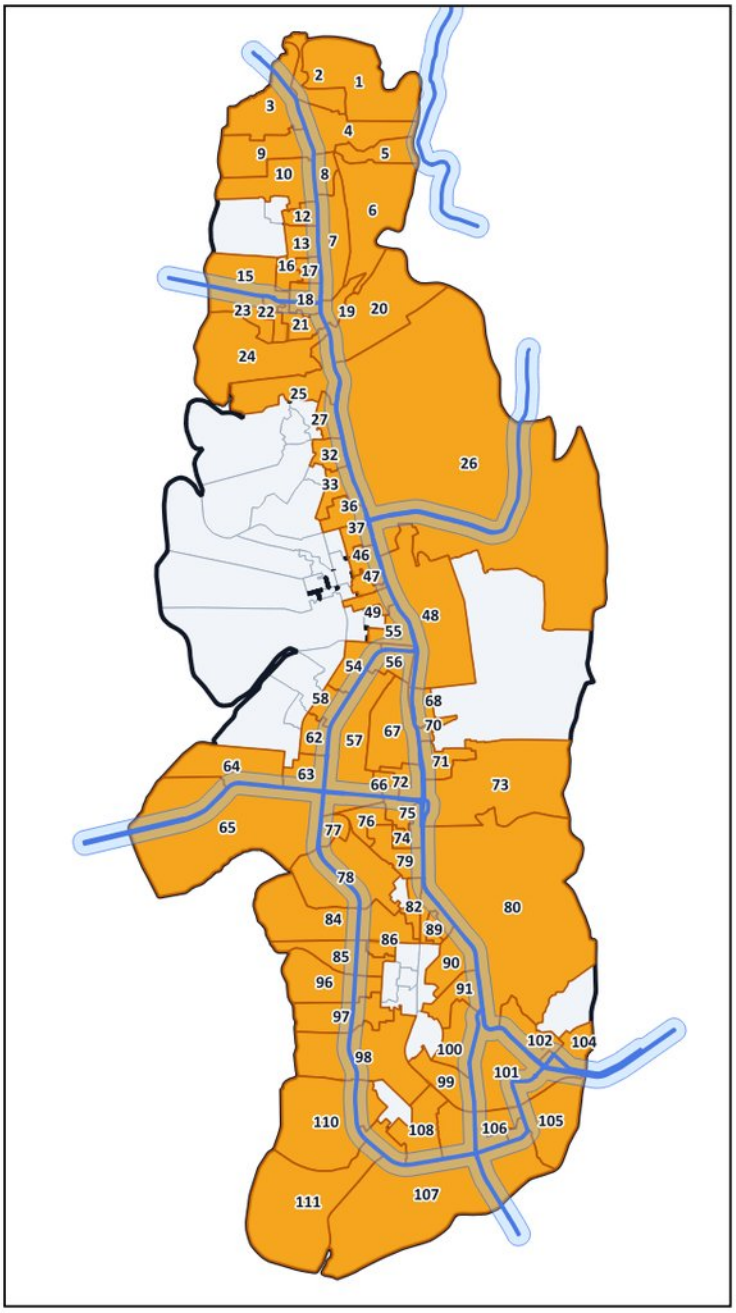


Table: first 12 of 78 rows, sorted by UCSI rank (1 = highest stress).

Ward_ID	BuiltUp_Pct	LST_C	UCSI	UCSI_Rank
70	98.25	47.597	4.48	1
57	84.13	45.847	4.48	2
67	80.85	46.442	4.43	3
66	86.84	45.302	4.39	6
89	95.2	45.716	4.31	7
3	47.19	44.653	4.29	9
16	69.18	45.07	4.27	10
75	86.71	45.565	4.27	11
8	70.96	45.733	4.24	12
72	75.3	46.346	4.21	13
22	78.05	44.055	4.2	15
12	86.91	45.074	4.19	17

HOW IT WAS RUN IN QGIS

```
Vector > Geoprocessing > Buffer
Input: major_roads (motorway, trunk, primary)
Distance: 200 m, dissolve
then Select by Location:
wards intersect the buffer
```

RESULT

78 of 111 wards

WHAT IT MEANS

78 wards have a motorway, trunk or primary road within 200 m (blue buffer). Two of these roads, Thane-Belapur Road and Sion-Panvel Highway, were reported waterlogged on 26 May 2025 (news report).

Result of the query

Other wards

0 2 4 km

Query S3: Hospitals inside the highest-stress wards

8 of 93 mapped hospitals

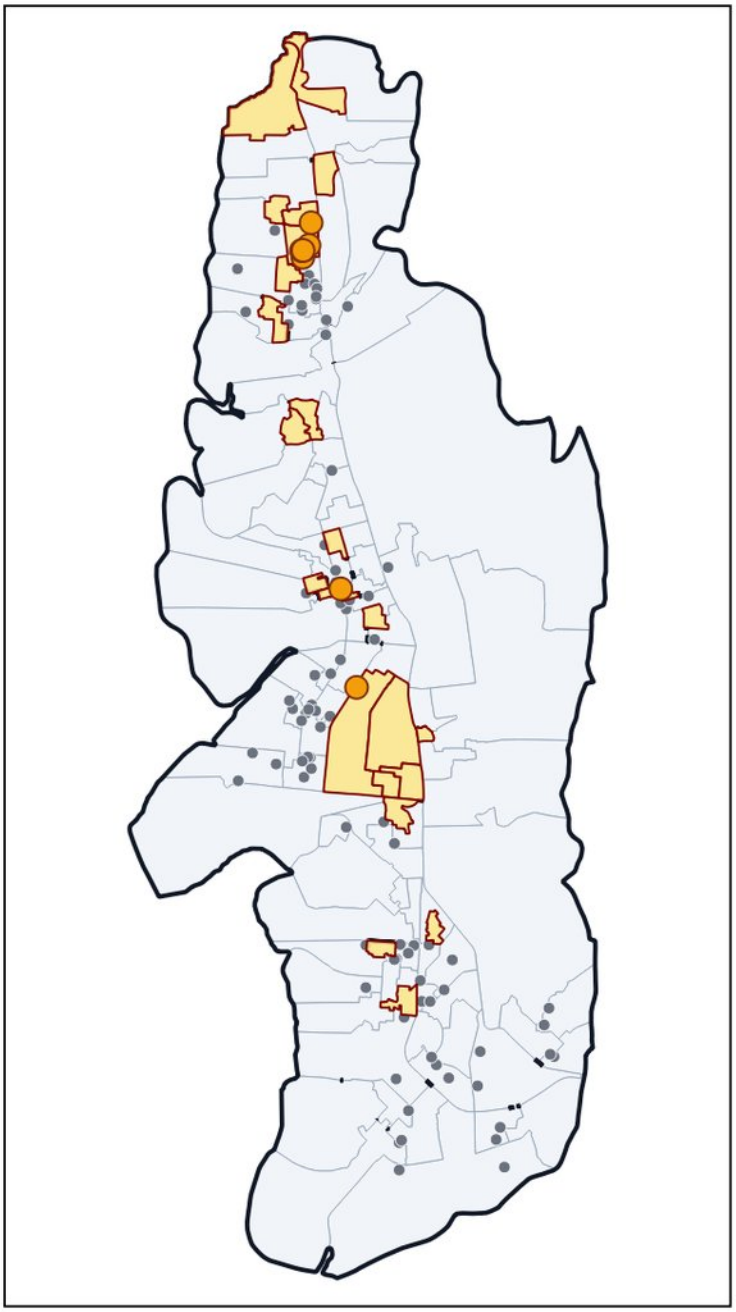


Table: all 8 rows, sorted by Ward_ID.

Hosp_ID	Name	Type	Ward_ID
38	Mody Hospital Iccu	hospital	12
6	Sai Seva Hospital and Icu, Mumbai	hospital	13
15	Indravati Hospital	hospital	13
50	Sanjivni Hospital, Thane	hospital	13
61	NMMC MCH Airoli	hospital	13
91	Unnamed hospital	hospital	13
44	Aakashdeep Hospital and I.C.U	hospital	44
71	Healing Hands Clinic - Vashi, New Mumbai	hospital	57

HOW IT WAS RUN IN QGIS

```
Step 1: Select by expression
  "UCSI_Class" = 'Highest'  (23 wards)
Step 2: Select by Location
  Input: hospitals
  Predicate: are within
  Comparison: the 23 wards
```

RESULT

8 of 93 mapped hospitals

WHAT IT MEANS

Only 8 of the 93 mapped hospitals and clinics are inside the 23 highest-stress wards (yellow). OpenStreetMap may be missing some facilities, so treat this as a first check.

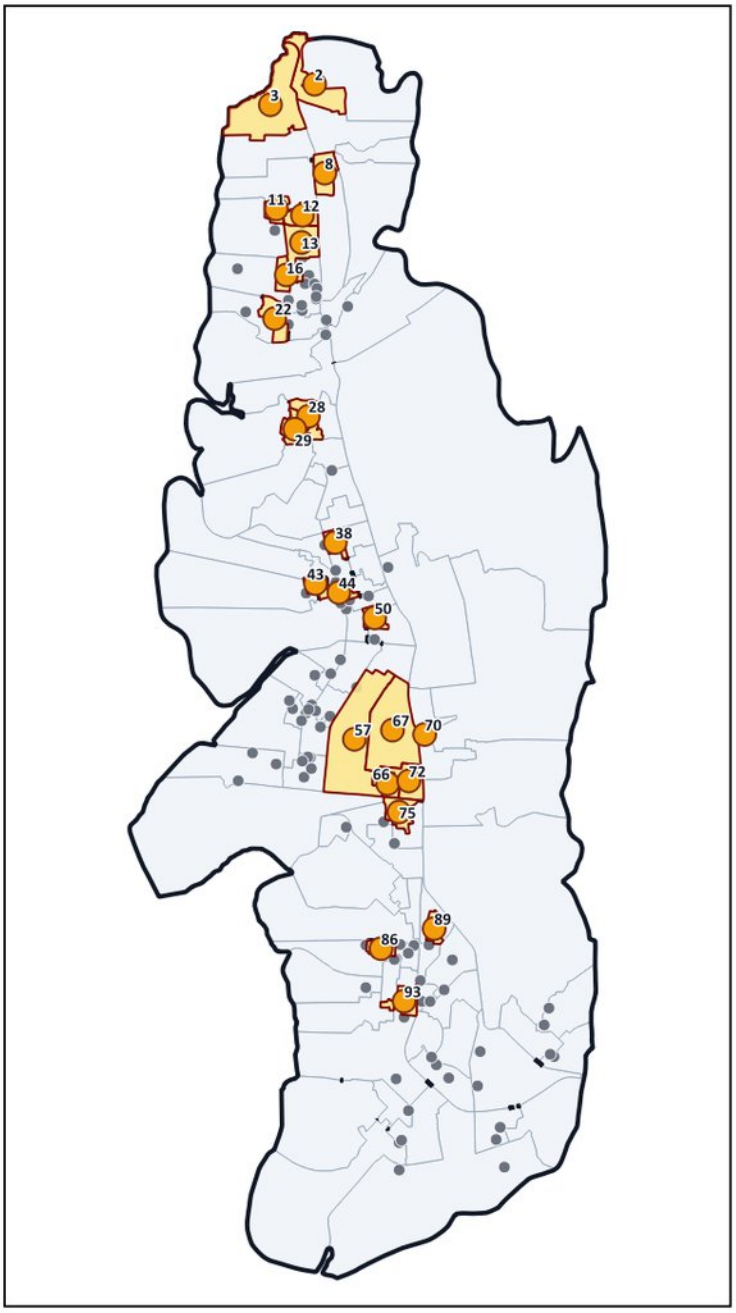
Result of the query

Other wards

0 2 4 km

Query S4: Nearest hospital to each highest-stress ward

mean 693 m, longest 2505 m



HOW IT WAS RUN IN QGIS

```
Vector > Geoprocessing > Centroids
  (the highest-stress wards)
Vector > Data Management >
  Join attributes by nearest
  Input: ward centroids
  Join layer: hospitals, 1 neighbour
Field Dist_m = round(distance), metres
```

RESULT

mean 693 m, longest 2505 m

WHAT IT MEANS

From the centre of each highest-stress ward, the nearest mapped hospital is 693 m away on average. The longest distance is 2505 m (ward 2). Distances are straight lines, not road distance.

Result of the query

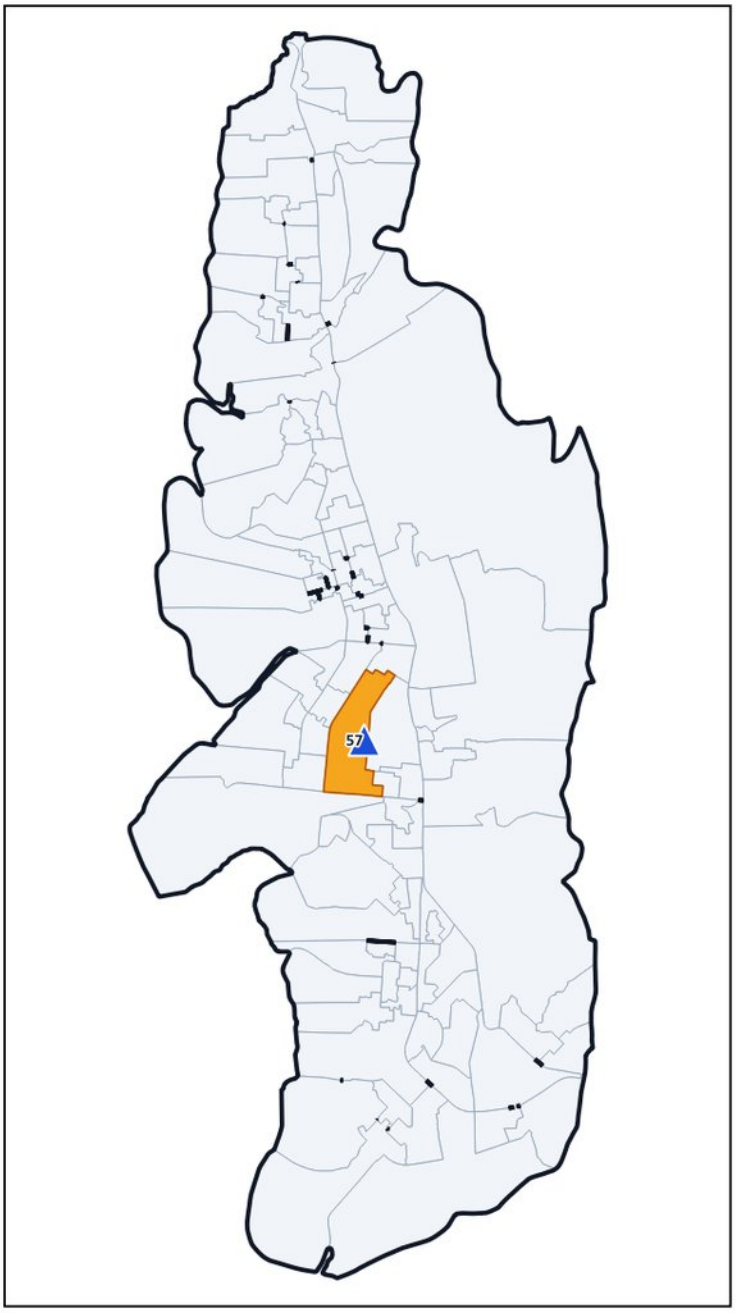
Other wards

Table: first 12 of 23 rows, sorted by distance, longest first.

Ward_ID	Hosp_Name	Dist_m	UCSI
2	Mody Hospital Iccu	2,505	4.15
3	Mody Hospital Iccu	2,257	4.29
70	Healing Hands Clinic - Vashi, New Mumbai	1,496	4.48
28	Divine Multispecialty Hospital and Research Centre	1,067	4.3
67	Healing Hands Clinic - Vashi, New Mumbai	1,009	4.43
29	Divine Multispecialty Hospital and Research Centre	1,003	4.4
8	Mody Hospital Iccu	930	4.24
72	Sanpada General Hopital	868	4.21
66	Sanpada General Hopital	694	4.39
57	Pkc Hospital	608	4.48
11	Avadhoot Hospital and Iccu	392	4.41
50	Unnamed hospital	390	4.18

Query S5: Ward that contains the waterlogged APMC market

Ward 57: flood score 4.57, UCSI 4.48 (Highest)



HOW IT WAS RUN IN QGIS

```
Vector > Research Tools > Select by Location
Input: wards
Predicate: contain
Comparison: flood_reports where
  Report_Type = 'Waterlogged site'
```

RESULT

Ward 57: flood score 4.57, UCSI 4.48 (Highest)

WHAT IT MEANS

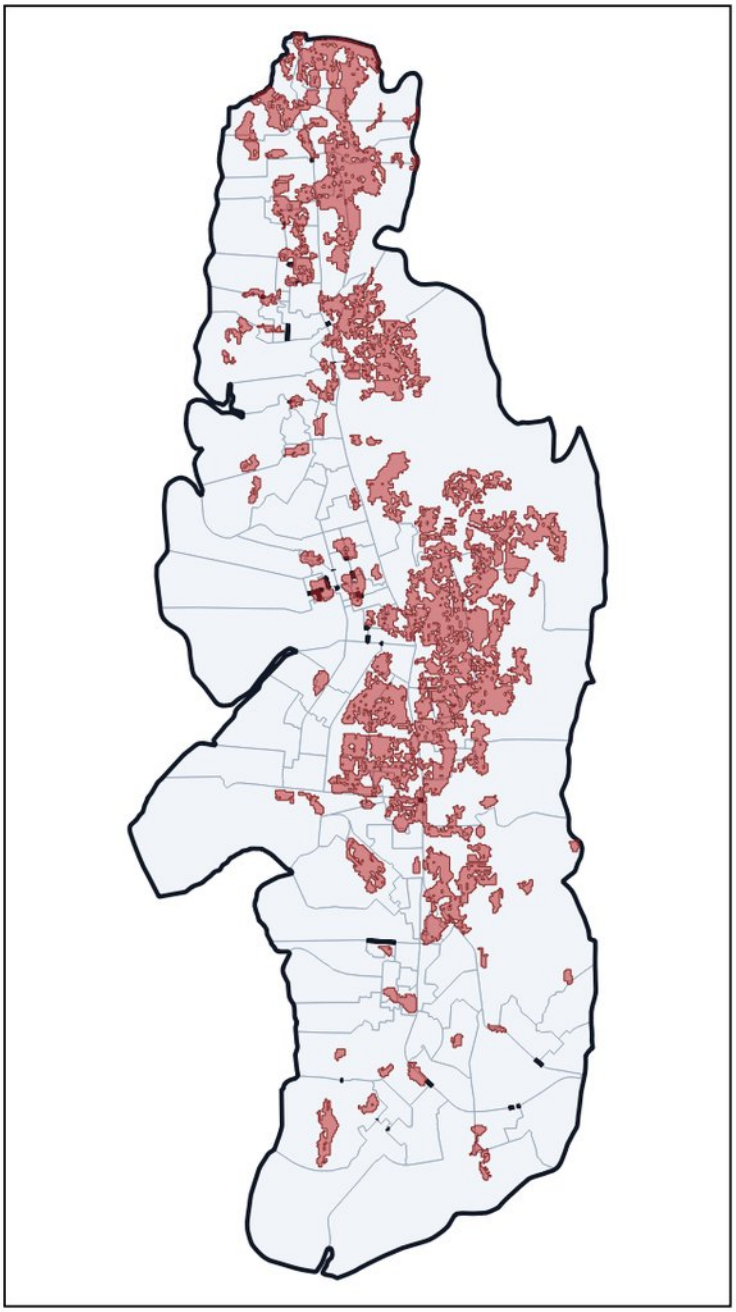
The APMC market site reported waterlogged on 26 May 2025 lies in ward 57. The model gives this ward a high flood score (4.57) and UCSI rank 2 of 111, so the news report and the model agree here. This is a small consistency check, not proof. The point is approximate (centre of the market yards in OpenStreetMap).

Table: all 1 rows, sorted by Ward_ID.

Ward_ID	Flood_Score	UCSI	UCSI_Class	UCSI_Rank
57	4.57	4.48	Highest	2

Query S6: Where hot and low-green land overlaps, ward by ward

17.6 km² in 71 patches; 85 wards touched



HOW IT WAS RUN IN QGIS

```
Raster calculator:
  LST >= 45 AND NDVI < 0.2
Polygonise, keep patches >= 2 ha
Vector > Geoprocessing > Intersection:
  wards with hot patches
Field calculator: $area / 10000 (ha)
Statistics by category: sum per ward
```

RESULT

17.6 km² in 71 patches; 85 wards touched

WHAT IT MEANS

Ward 26 has the largest hot and low-green area (281 ha). Among wards larger than 50 ha, ward 67 has the highest share (60.3% of the ward). These are the first places to look at for trees or cool roofs.

Hot and low-green patch (2 ha or more)

Other wards

0 2 4 km

Table: first 12 of 85 rows, sorted by area, largest first.

Ward_ID	Hot_lowgreen_ha	Ward_area_ha	Share_pct
26	281.5	1,686.2	16.7
69	191	647.1	29.5
48	147.1	290.8	50.6
80	107.3	802	13.4
1	82	201.7	40.6
57	80.2	155.5	51.5
67	71.2	118.1	60.3
20	60.8	194.5	31.2
6	56.5	195.1	29
3	44.1	123.4	35.8
7	41.5	99.7	41.7
73	35.1	326.7	10.7